Evaluating Membership and Broadcast Protocols

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Abstract

Gossip-based protocols have emerged as an effective approach to implement scalable and resilient broadcast mechanisms in distributed systems. In such systems, nodes rely on membership protocols to maintain partial views of the network, which directly influence the efficiency and reliability of message dissemination. The performance of gossip-based broadcast depends on the interaction between membership and broadcast strategies, particularly under varying system conditions.

Several approaches can be used to construct and maintain these overlays, each offering different trade-offs in terms of reliability, latency, and communication overhead. In this paper, we implement and experimentally evaluate combinations of membership protocols, namely Cyclon and HyParView, with broadcast protocols such as Flood and Eager-Push Gossip, as well as a custom-designed gossip-based solution. Our results show that different protocol combinations exhibit distinct performance characteristics, highlighting trade-offs between reliability and latency, and demonstrating that the choice of membership protocol plays a key role in determining the effectiveness of broadcast mechanisms.

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1 Introduction

Modern distributed systems often need to send a message to all nodes participating in the system, an operation known as broadcast. The simplest way to do this is flooding, where every node that receives a message immediately forwards it to all of its neighbors. This guarantees that every node will eventually receive the message, but it comes at a steep cost. Because every node forwards to every neighbor, the total number of messages in the network grows very quickly as the system scales, filling up network links and forcing each node to spend time processing and discarding large numbers of duplicate messages. This extra work does not go unnoticed: the congestion and processing overhead introduced by flooding ultimately manifests as increased end-to-end latency, meaning messages take longer to arrive not because the network is slow, but because it is overloaded.

A more efficient alternative is gossip-based broadcast, where instead of forwarding to every neighbor, each node picks only a small random subset of neighbors to forward to. The size of this subset is controlled by a parameter called the fanout. At first glance it might seem that reducing the number of forwarding targets would significantly hurt reliability, but in practice, if the fanout is well calibrated, the message still reaches the vast majority of nodes in the system. The core argument is therefore that the overhead flooding introduces does not meaningfully improve reliability over a well tuned gossip protocol, it only adds congestion and latency.

For gossip to work well, nodes need a way to know which other nodes exist in the system and to maintain a set of neighbors to gossip with. This is the role of an overlay network protocol. In this work we consider two such protocols: Cyclon and HyParView. Cyclon maintains a single list of neighbors and periodically shuffles it with other nodes to keep the view random and fresh. It is simple and works well in stable conditions. HyParView takes a different approach and maintains two lists: an active view of neighbors that are currently in use, and a passive view of backup neighbors that are kept in reserve. When an active neighbor leaves or stops responding, HyParView can immediately replace it with a node from the passive view without waiting for the next maintenance cycle. This makes HyParView more resilient when nodes frequently join and leave the network, a condition known as churn. In scenarios with little churn however, the simpler design of Cyclon is expected to introduce slightly less overhead and therefore slightly lower latency.

On top of these overlay protocols we implement a hybrid gossip broadcast protocol. Rather than always forwarding the full message to neighbors, our protocol splits neighbors into two groups per message. A small subset called eager targets receives the full message right away. Another subset called lazy targets receives only a small announcement indicating that the sender has a message available. If a lazy target has not yet seen that message, it can request the full content, effectively pulling it on demand. This reduces unnecessary full message transmissions when nodes have already received the message through another path. Additionally, our protocol monitors how many duplicate messages it is receiving and uses this information to dynamically adjust the eager fanout over time, growing it when too few nodes seem to be receiving messages and shrinking it when too many duplicates are observed. We expect this adaptive behavior to reduce message overhead compared to a static eager push approach, though the dynamic adjustment of the fanout may occasionally cause short periods of reduced delivery reliability.

In this paper we experimentally study these three aspects: whether flooding truly introduces more latency than gossip for similar reliability levels, whether HyParView outperforms Cyclon under high churn while Cyclon holds an edge in stable networks, and whether our hybrid adaptive protocol improves on static gossip in terms of

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overhead while remaining competitive in reliability. The remainder of this paper is organized as follows. Section 2 discusses related work on overlay networks and gossip protocols. Section 3 describes our implementation in detail. Section 4 presents and discusses our experimental results. Finally, Section 5 concludes the paper.

2 Related Work

This section reviews prior work that is necessary to contextualize the protocol combinations studied in this paper. We first discuss random overlay construction protocols, then gossip-based broadcast protocols, and finally prior experimental studies that compare dissemination strategies in similar settings.

2.1 Random Overlay Networks

Random overlay networks are a common building block for scalable distributed systems because they provide each node with only a partial view of the network while still preserving good connectivity and short communication paths. Instead of requiring every node to know all other participants, these protocols maintain a small and continuously refreshed neighbor set, which is enough to support robust dissemination and membership management at scale.

Cyclon is one of the best-known peer-sampling protocols in this space. Its main goal is to maintain a continuously evolving random overlay by periodically exchanging small neighbor samples between nodes. This periodic shuffling process keeps views fresh, removes stale entries, and helps preserve overlay randomness over time. Because of its simple design, Cyclon is often used as a baseline membership mechanism in experimental studies of gossip-based systems.

HyParView was proposed to improve robustness under churn while still maintaining a lightweight overlay. In contrast with protocols that rely on a single partial view, HyParView separates neighbor management into two structures: a small active view used for communication and a larger passive view used as backup. This distinction allows the protocol to react quickly to failures by replacing lost active neighbors from the passive view, while periodic maintenance keeps the passive view updated. As a result, HyParView is generally regarded as better suited to highly dynamic environments.

These two protocols represent different design choices in overlay construction. Cyclon emphasizes simplicity and randomness through periodic exchanges, whereas HyParView combines reactive and periodic mechanisms to improve failure resilience. This makes them natural candidates for comparison when studying how the choice of membership layer affects broadcast performance.

2.2 Gossip-Based Broadcast

Gossip-based broadcast protocols were introduced as a scalable alternative to deterministic flooding. In flooding, each node forwards a received message to all its neighbors. This strategy is simple and achieves full dissemination in connected overlays, but it generates a large number of duplicate transmissions and places unnecessary load on the network. As the system grows, this redundancy increases overhead and may also increase end-to-end latency due to congestion and repeated processing.

Eager-push gossip reduces this cost by forwarding each message only to a subset of neighbors, usually chosen at random. The size of this subset, commonly called the fanout, determines the trade-off between reliability and overhead. When the fanout is too small, messages may fail to reach the whole system. When it is too large, the protocol starts behaving closer to flooding and produces many duplicates. A well-chosen fanout therefore allows gossip-based broadcast to retain high coverage while significantly reducing communication costs.

Several works also explore hybrid dissemination strategies. Instead of always sending full payloads, some protocols first advertise the existence of a message and only transmit the full content when needed. This idea appears in lazy-push and pull-based designs, where message identifiers are disseminated separately from message bodies. Such approaches can reduce redundant full-message transmissions, especially when many nodes have already received the same content through different paths.

Plumtree is one of the best-known protocols combining eager and lazy strategies. Its design attempts to exploit the low latency of tree-based dissemination while preserving the resilience of gossip through a backup mechanism. This line of work shows that combining dissemination modes can provide better trade-offs than purely eager or purely lazy schemes.

Our hybrid protocol is inspired by this family of approaches. However, instead of fixing the balance between eager and lazy dissemination once and for all, we adapt the eager fanout dynamically according to the observed duplication rate. This design choice aims to let the protocol react to the conditions of the overlay rather than relying on a single static parameter.

2.3 Experimental Studies of Gossip Protocols

A substantial body of prior work evaluates gossip-based dissemination protocols experimentally, typically focusing on metrics such as reliability, dissemination latency, message overhead, and robustness under churn. These studies repeatedly show that dissemination behavior cannot be analyzed independently from overlay construction. The quality of the membership layer influences path diversity, redundancy, and the ability to recover from failures, all of which affect broadcast performance.

Prior experimental comparisons also show that flooding tends to achieve high reliability at the cost of excessive overhead, whereas probabilistic gossip often achieves comparable coverage with fewer transmissions when its parameters are well tuned. Similarly, overlays designed for dynamic environments tend to outperform simpler overlays under churn, even if they may introduce some additional maintenance cost in stable scenarios.

The goal of our work is aligned with this experimental tradition. Rather than proposing a purely theoretical argument, we implement different combinations of membership and broadcast protocols and analyze their behavior under common dissemination metrics. In this sense, our study is not only about the protocols in isolation, but also about the interaction between overlay maintenance and message dissemination.

3 Implementation

This section describes the implementation of the system developed for this project. The goal of the implementation was to make it possible to experimentally compare multiple combinations of membership and broadcast protocols within a common execution framework. To achieve this, we organized the system into two main layers: a membership layer, which is responsible for maintaining the overlay network, and a broadcast layer, which uses that overlay to disseminate messages. We first present the overall organization of the implementation, then describe the membership protocols, and finally discuss the broadcast protocols and the custom hybrid solution.

3.1 System Organization

The system was implemented using the Babel framework, which supports the composition of protocols through message handlers, timers, channel events, and inter-protocol notifications. This model was particularly useful for the project because it allowed us to separate membership logic from broadcast logic while still enabling communication between both layers.

The membership protocols expose changes in the local neighborhood through notifications that inform the broadcast layer when neighbors become available or unavailable. This design keeps the broadcast protocols independent from the internal details of overlay maintenance and makes it possible to combine the same broadcast algorithm with different membership mechanisms.

At a high level, every node executes one membership protocol and one broadcast protocol. The membership protocol builds and maintains the local partial view of neighbors. The broadcast protocol relies on that view to forward messages. This separation mirrors the conceptual structure of the project and simplifies both experimentation and code reuse.

3.2 Membership Layer

3.2.1 Cyclon. Our Cyclon implementation follows the standard peer-sampling approach based on partial views and periodic shuffling. Each node maintains a view containing neighbor entries associated with an age. At regular intervals, the node increments the age of its entries, selects one neighbor, and exchanges a sample of entries with that node. This process gradually refreshes the view and helps preserve randomness in the overlay.

The main advantage of Cyclon in our implementation is its simplicity. Since all view maintenance is driven by periodic exchanges, the protocol is compact and easy to integrate with the rest of the system. However, this same design means that recovery from failures is less immediate, because failed nodes are only removed or replaced when subsequent maintenance actions occur.

3.2.2 HyParView. Our HyParView implementation extends the simpler peer-sampling approach by maintaining two views: an active view and a passive view. The active view contains the neighbors currently used for communication, while the passive view contains backup nodes that can be promoted when necessary.

When a node joins the system, it contacts a known participant and sends a join request. The contacted node inserts the newcomer into its active view and propagates a `FORWARDJOIN` message through the overlay using a random walk controlled by a time-to-live parameter. Intermediate nodes may insert the new node into their passive view, while the final node in the walk inserts it into its active view. This mechanism helps distribute knowledge of the joining node across the network while keeping active views small.

When an active connection fails, the protocol immediately tries to replace the failed node using an entry from the passive view. This is one of the central design goals of HyParView and the main reason it is expected to perform better under churn. The passive view itself is refreshed periodically through shuffle messages that exchange neighbor information across the overlay.

3.2.3 Why Compare Cyclon and HyParView. These two membership protocols were implemented because they represent different philosophies of overlay maintenance. Cyclon relies on a single randomized structure maintained through periodic exchanges, while HyParView relies on a small communication view plus a larger backup view, combining reactive and periodic mechanisms. By implementing both, we can observe how broadcast protocols behave when the overlay is either simpler and lighter or more explicitly designed for fault recovery.

3.3 Broadcast Layer

3.3.1 Flood Broadcast. The flood broadcast implementation is the simplest dissemination mechanism in the system. When a node first receives a message, it delivers it locally and forwards it to all current neighbors except, when applicable, the sender from which the message was received. To avoid infinite retransmission loops, each node stores the identifiers of messages it has already processed and discards duplicates.

This protocol serves as a baseline. Its expected strength is very high reliability in connected overlays. Its expected weakness is the large number of duplicate transmissions generated by forwarding to all neighbors.

3.3.2 Eager-Push Gossip. In eager-push gossip, a node forwards each newly received message only to a randomly selected subset of its neighbors. The size of this subset is the fanout. As in flood broadcast, duplicate messages are ignored through the use of message identifiers.

Compared with flooding, this protocol introduces an explicit trade-off. Forwarding to fewer neighbors reduces overhead, but also makes dissemination depend more strongly on the quality of the overlay and on the selected fanout. This protocol is therefore useful to study the cost of probabilistic dissemination under different membership layers.

3.3.3 Hybrid Adaptive Gossip. The custom protocol developed for this project combines eager and lazy dissemination. When a node first receives a message, it divides the available neighbors into two groups. A subset of neighbors receives the full message immediately and forms the eager set. The remaining selected neighbors receive only a lightweight notification containing the message identifier and form the lazy set.

If a node receives a lazy notification for a message it does not yet have, it sends an explicit request to obtain the full payload. This design avoids sending the full message to every target immediately

and reduces unnecessary transmissions when the message has already reached a node through another path.

The protocol also includes an adaptive mechanism based on duplicate observations. Each node tracks how many first-time and duplicate full messages it receives over a period of time. When the observed duplicate rate is high, the eager fanout is reduced. When the duplicate rate is low, the eager fanout is increased, within predefined bounds. The goal is to adapt dissemination aggressiveness to the redundancy currently observed in the network.

Algorithm 1 Hybrid Gossip Broadcast

```
State:
neighbours
received
requested
messageCache
currentEagerFanout
lazyFanout
minEagerFanout
maxEagerFanout
adaptWindow
newCount ← 0
duplicateCount ← 0
procedure BroadcastRequest(mid, sender, payload)
    m ← (mid, sender, payload)
    ReceiveFull(m, self)
end procedure
procedure ReceiveFull(m, from)
    if m.mid ∉ received then
        received ← received ∪ {m.mid}
        messageCache[m.mid] ← m
        newCount ← newCount + 1
        Deliver message to application
        eagerTargets ← random subset of neighbours excluding from
        lazyTargets ← random subset excluding eagerTargets
        for all p ∈ eagerTargets do
            Send FULL message to p
        end for
        for all p ∈ lazyTargets do
            Send IHAVE(m.mid) to p
        end for
    else
        duplicateCount ← duplicateCount + 1
    end if
    AdaptFanout
end procedure
procedure ReceiveIHave(mid, from)
    if mid ∉ received and mid ∉ requested then
        requested ← requested ∪ {mid}
        Send REQUEST(mid) to from
    end if
end procedure
procedure ReceiveRequest(mid, from)
    if mid ∈ messageCache then
        Send FULL(mid) to from
    end if
end procedure
procedure AdaptFanout
    total ← newCount + duplicateCount
    if total < adaptWindow then
        return
    end if
    dupRate ← duplicateCount/total
    if dupRate > 0.5 then
        decrease eager fanout
    else if dupRate < 0.2 then
        increase eager fanout
    end if
    reset counters
end procedure
```

3.4 Correctness Argument

The protocol preserves *at-most-once delivery* because each node keeps a local set of previously received message identifiers. A message is only delivered to the application the first time its identifier is observed. Any subsequent reception of the same full message is classified as a duplicate and is ignored for delivery purposes.

Assuming the overlay remains connected and messages between correct nodes are eventually delivered, the protocol also ensures *eventual delivery* to nodes that are reachable from the broadcaster. This follows from the fact that, whenever a node receives a message for the first time, it propagates either the full message to eager targets or an `IHAVE` announcement to lazy targets. A node that learns about a message through an `IHAVE` and does not yet possess the payload explicitly requests it through a `REQUEST` message. Therefore, lazy dissemination does not suppress delivery; it only postpones full payload transmission until it is needed.

The adaptive fanout mechanism does not affect safety, since it only changes the number of eager targets selected in future transmissions and does not bypass deduplication or local delivery rules. Its impact is therefore on performance and reliability under particular parameter choices, but not on the at-most-once delivery property.

3.5 Implementation Summary

In summary, the implementation was structured to support a clean comparison between overlay construction and dissemination strategies. Cyclon and HyParView provide two distinct membership mechanisms, while flooding, eager-push gossip, and the proposed hybrid adaptive protocol provide three dissemination strategies with different reliability and overhead trade-offs.

4 Experimental Evaluation

This section evaluates the performance of different combinations of membership and broadcast protocols. We first describe the experimental setup and methodology, and then present and discuss the results in terms of reliability, latency, and communication overhead.

4.1 Methodology

All experiments were conducted using the Babel framework, where each node executes one membership protocol and one broadcast protocol. We evaluate different combinations of:

- Membership protocols: Cyclon and HyParView
- Broadcast protocols: Flood, Eager-Push Gossip, and Hybrid Adaptive Gossip

Each experiment consists of multiple broadcast rounds where messages are periodically injected into the system. Nodes log message reception events, which are later processed offline to compute performance metrics.

To ensure statistical significance, each configuration is executed multiple times and results are aggregated across runs.

4.1.1 Metrics. We evaluate the system using the following metrics:

- **Reliability**: percentage of nodes that successfully receive a message.
- **Latency**: time elapsed between message creation and its reception at each node.
- **Message Overhead**: total number of messages exchanged during execution.

4.1.2 System Parameters. All experiments were conducted using the same base configuration unless otherwise stated.

Membership Parameters. For Cyclon, we use the following configuration:

- View size ($maxN$): 8
- Shuffle period: 2000 ms
- Shuffle subset size: 4

For HyParView, the parameters are:

- Active view size: 5
- Passive view size: 20
- Active Random Walk Length (ARWL): 5
- Passive Random Walk Length (PRWL): 2

Additionally, the passive view maintenance is configured as:

- Shuffle TTL: 3
- Shuffle period: 4000 ms
- Active nodes exchanged (k_{active}): 3
- Passive nodes exchanged ($k_{passive}$): 4

Broadcast Parameters. For Eager-Push Gossip:

- Fanout: 5

For the Hybrid Adaptive Gossip protocol:

- Initial eager fanout: 3
- Lazy fanout: 2
- Minimum eager fanout: 2
- Maximum eager fanout: 5
- Adaptation window: 12 messages

Application Parameters.

- Payload size: 20 bytes
- Broadcast interval: 3000 ms
- Operation pool size: 1000 messages

4.1.3 Discussion of Parameters. The chosen parameters reflect typical configurations used in gossip-based systems. In Cyclon, the view size and shuffle parameters aim to maintain a sufficiently random overlay while keeping overhead low. In HyParView, the small active view combined with a larger passive view enables fast recovery from failures while preserving scalability.

For dissemination, the eager-push fanout of 5 represents a moderately aggressive gossip strategy, expected to provide high reliability. The hybrid protocol starts with a lower eager fanout and adapts dynamically within predefined bounds, allowing it to react to observed duplication levels in the network.

These parameter choices aim to balance reliability, latency, and overhead, while allowing meaningful comparison between protocol combinations.

4.2 Results

In this section, we present the results obtained for each combination of membership and broadcast protocols. We analyze reliability, latency, and message overhead based on the collected execution logs.

Table 1: Reliability results for each protocol combination.

Configuration	Avg (%)	Min (%)	Max (%)
EagerPush + Cyclon	99.08	91	100
EagerPush + Full	99.43	93	100
EagerPush + HyParView	100.0	100	100
Flood + Cyclon	99.99	97	100
Flood + Full	100.0	100	100
Flood + HyParView	98.02	1	99
Hybrid + Full	98.27	90	100
Hybrid + Cyclon	87.96	65	99
Hybrid + HyParView	97.31	1	99

4.2.1 Reliability. Table 1 summarizes the reliability results.

Flood and Eager-Push approaches generally achieve very high reliability, with several configurations reaching near-perfect delivery. Notably, EagerPush combined with HyParView achieves 100% reliability across all runs.

However, the hybrid protocol exhibits more variability, particularly when combined with Cyclon, where reliability drops to approximately 88% on average. This suggests that the adaptive mechanism may be too conservative in some scenarios, reducing message propagation.

4.2.2 Latency. Table 2 presents the latency results.

Table 2: Latency results for each protocol combination (ms).

Configuration	Avg	Min	Max
EagerPush + Cyclon	6.13	3	706
EagerPush + Full	6.14	3	749
EagerPush + HyParView	4.57	2	698
Flood + Cyclon	66.74	3	862
Flood + Full	198.87	20	13563
Flood + HyParView	3.29	0	204
Hybrid + Full	7.03	4	475
Hybrid + Cyclon	7.65	3	642
Hybrid + HyParView	5.52	0	815

Gossip-based approaches (EagerPush and Hybrid) consistently achieve low latency, typically between 4 and 7 ms. EagerPush combined with HyParView achieves the lowest latency among these configurations.

Flood exhibits significantly higher latency, especially when combined with Full Membership, where the average latency reaches nearly 200 ms. This is likely due to excessive message redundancy and network congestion.

Interestingly, Flood combined with HyParView shows very low latency, suggesting that the smaller active view reduces congestion effects.

4.2.3 Message Overhead. The total number of messages exchanged across all configurations is approximately 239k–240k messages.

Although Flood is expected to generate the highest overhead, the results indicate that overhead differences are relatively small across configurations. This is likely due to the fixed duration and workload of the experiments.

Nevertheless, Flood-based approaches still produce more redundant message transmissions, which contributes to increased latency.

4.2.4 Membership Overhead. Table 3 summarizes the membership overhead results in terms of control messages per second.

Table 3: Membership overhead (messages per second).

Configuration	Overhead (msg/s)
EagerPush + Cyclon	115.73
Hybrid + Cyclon	115.18
Flood + Cyclon	—
EagerPush + HyParView	98.48
Flood + HyParView	101.65
Hybrid + HyParView	97.51

The results show that Cyclon incurs higher membership overhead than HyParView, with values around 115 msg/s compared to approximately 98–102 msg/s for HyParView. This difference is expected due to the more frequent shuffle operations in Cyclon (every 2 seconds), while HyParView performs maintenance less frequently (every 4 seconds).

Additionally, the membership overhead remains consistent across different broadcast protocols. This confirms that overlay maintenance is largely independent from the dissemination strategy, as expected from the modular design of the system.

The missing value for Flood + Cyclon is due to incomplete log data and is therefore omitted from the analysis.

4.3 Discussion

The results highlight several important trade-offs.

First, Eager-Push Gossip achieves an excellent balance between reliability and latency. In particular, when combined with HyParView, it reaches perfect reliability while maintaining the lowest latency among all configurations.

Flood broadcast guarantees high reliability but at the cost of significantly higher latency, especially in dense overlays such as Full Membership. This confirms that excessive redundancy leads to network congestion.

The hybrid adaptive protocol reduces latency compared to Flood and maintains reasonable reliability. However, its performance is sensitive to the choice of membership protocol. In particular, when combined with Cyclon, reliability drops noticeably, indicating that the adaptive fanout may be too aggressive in reducing eager transmissions.

Finally, the choice of membership protocol has a clear impact. HyParView generally provides better reliability and lower latency, likely due to its structured active view and fast recovery mechanisms. Cyclon, while simpler, may lead to less stable dissemination paths, especially for adaptive protocols.

5 Conclusions

In this paper, we evaluated the interaction between membership protocols and broadcast strategies in gossip-based distributed systems. We studied Cyclon and HyParView overlays combined with Flood, Eager-Push Gossip, and a Hybrid Adaptive Gossip protocol.

Our results show that Eager-Push Gossip provides the best overall trade-off between reliability and latency. In particular, when combined with HyParView, it achieves perfect reliability with very low latency, demonstrating the effectiveness of combining structured overlays with controlled gossip dissemination.

Flood broadcast guarantees high delivery ratios but introduces significant latency due to message redundancy and network congestion, especially in fully connected overlays.

The hybrid adaptive protocol shows promising results in reducing overhead while maintaining acceptable latency. However, its reliability is sensitive to the underlying membership protocol, suggesting that further tuning of the adaptation mechanism is needed.

Overall, the results confirm that both the broadcast strategy and the membership protocol play a crucial role in system performance, and that careful parameter tuning is essential to achieve an optimal balance.

5.1 Limitations

This evaluation was conducted under a fixed set of parameters and stable network conditions. The behavior of the protocols under high churn or network failures was not explored.

5.2 Future Work

Future work could explore more advanced adaptive strategies, including dynamic tuning based on real-time network feedback. It would also be interesting to evaluate these protocols under more challenging conditions, such as high churn or partial network failures, and to compare them with other state-of-the-art gossip-based dissemination techniques.